# FEASIBILITY REPORT

**DataTalk:** Automated Data Cleaning and Natural Language Query Engine

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## ABSTRACT

DataTalk is an end-to-end data analysis platform that democratizes data accessibility by automating both data cleaning and natural language query execution. This updated feasibility report evaluates the technical, economic, operational, legal, and schedule feasibility of developing DataTalk as a Final Year Project, incorporating significant enhancements to data quality assessment, multi-source data handling, and impact analysis. The system integrates three core modules:
(1) CleanQuery, an automated data cleaning engine with advanced data profiling and quality scoring
(2) SourceGuard, a multi-source data consolidation system with conflict detection and reliability assessment
(3) InsightFlow, a schema-aware natural language to SQL converter that eliminates hallucination through strict schema injection and validation.
New capabilities include data noise detection with quality scores, root cause analysis of data issues, before/after change tracking, inter-source conflict detection and resolution, impact analysis on downstream analytics, and an interactive insights generation dashboard. The project is technically feasible using Python (Flask/FastAPI backend), PostgreSQL (database), OpenAI GPT-4 API (for NLP), and Streamlit (frontend). Economically, the project requires minimal investment (0–15,000 PKR) as all core infrastructure uses open-source or free-tier SaaS. The 6–7-month development timeline is operationally feasible for a solo developer with data engineering and Python competencies. This report concludes that DataTalk is highly feasible and recommends proceeding with full development, with particular focus on robustness of the data cleaning pipeline, multi-source reconciliation, and comprehensive error handling for non-technical users.

# 1. INTRODUCTION

## 1.1 Background and Problem Context

According to industry research, data professionals spend approximately 80% of their working time on data cleaning and preparation before any actual analysis can begin [1]. This statistic, derived from CrowdFlower's 2016 survey of data scientists, remains consistent today. The barrier to data accessibility exists at three distinct levels:
**Data Quality:** Raw data arrives with missing values, inconsistent formats, duplicate records, outliers, and inconsistencies. Organizations often lack visibility in data quality before analysis begins.
**Query Complexity:** Extracting insights from structured databases requires SQL knowledge that most domain experts do not possess.
**Multi-Source Integration:** Organizations frequently maintain data across multiple systems (databases, Excel files, APIs, data warehouses). Reconciling conflicts and assessing reliability across sources is manual, error-prone, and time-consuming.
Current solutions address these challenges in isolation and lack transparency on data quality issues. Spreadsheet tools like Microsoft Excel (used by over 750 million users [2]) require entirely manual data cleaning and offer no quality metrics. Enterprise BI platforms such as Tableau and Power BI assume data is already clean and structured. Text-to-SQL systems from major cloud providers (Amazon Athena, Snowflake, Databricks) are schema-aware but priced for large organizations at 50,000+ PKR/month [3], making them inaccessible to individual researchers, small businesses, and academic institutions. General-purpose language models can generate SQL but frequently hallucinate column names that do not exist in the actual database

[4], a critical failure mode for production systems. None of these solutions detect and quantify data quality issues, identify root causes, track changes transparently, handle multi-source conflicts, or explain how data quality issues impact downstream analysis.

## 1.2 Problem Statement

- No integrated, accessible tool currently combines advanced data quality profiling, automated preprocessing, multi-source conflict detection, and schema-aware natural language querying with impact analysis. The gap creates unnecessary friction:
- Non-technical users cannot assess data quality or understand data issues before analysis.
- No tools identify root causes of data problems (e.g., missing values originating from a specific source or column).
- Multi-source data integration requires manual conflict resolution with no reliability scoring.
- Users lack transparency on cleaning operations, what changed, how many records were affected, and why.
- No visibility into how data quality issues impact analysis outcomes (e.g., duplicates inflating revenue).
- Non-technical users cannot query databases without SQL knowledge.
- Existing solutions solve only one problem in isolation.
- Enterprise solutions are prohibitively expensive for individuals and SMEs.
- Open-source alternatives lack robustness, error handling, and multi-source support.
- Each new dataset requires manual cleaning and re-setup

## 1.3 Project Objectives

- Support data ingestion from multiple sources simultaneously (databases, DB connections, Excel, CSV, JSON)
- Implement comprehensive data profiling and quality scoring (0-100) for each dataset.
- Detect and identify 15+ types of data quality issues with precise location information (row, column, cell).
- Provide root cause analysis for data problems (e.g., 'Missing values in Age column originate from legacy database export').
- Assess reliability and quality scores for each data source.
- Detect conflicts between sources and recommend resolution strategies.
- Implement transparent change tracking, showing exactly what was changed, by how much, and why.
- Display before/after cleaning statistics (rows removed, cells fixed, types of changes).
- Implement automated data cleaning pipeline handling 15+ common data quality issues.
- Provide impact analysis explaining how data issues affect downstream analytics.
- Implement schema-aware natural language to SQL conversion with hallucination prevention.
- Provide interactive insights dashboard with natural language prompts and auto-generated visualizations.
- Create a user-friendly web interface for non-technical users.
- Ensure data security with read-only query execution and destructive command blocking.
- Achieve sub-3-second query response time for datasets up to 100K rows

## 1.4 Scope of Feasibility Study

This feasibility study examines DataTalk across five dimensions: technical feasibility (proven technology stack, architecture feasibility, complexity estimation), economic feasibility (minimal cost via free-tier SaaS and open-source tools), operational feasibility (solo developer capability, timeline realism), legal feasibility (data privacy, no regulatory barriers for academic FYP), and schedule feasibility (6–7-month development). The analysis includes multi-source data handling, quality assessment, and conflict detection. The analysis excludes enterprise-grade deployment, HIPAA/GDPR compliance for production use, and scalability beyond 10M rows. MVP scope focuses on single-table analysis; multi-table joins are Phase 2. Advanced predictive analytics and automated remediation are Phase 3.

# 2. FEASIBILITY ANALYSIS

## 2.1 Technical Feasibility

### 2.1.1 Recommended Technology Stack

The recommended stack prioritizes open-source libraries and free-tier SaaS for cost-effectiveness while maintaining production-grade reliability. Key additions include Great Expectations for data profiling and Pandas Profiling for statistical quality assessment.

### 2.1.2 Enhanced Architecture Design: Six-Stage Pipeline

DataTalk operates as an enhanced six-stage pipeline incorporating multi-source data handling, quality assessment, and conflict detection:

**Stage 1 - Multi-Source Data Ingestion:** CSV, Excel, JSON, direct database connection (MySQL, PostgreSQL, SQL Server), or API endpoints. File validation with automatic encoding detection and delimiter inference. Support for simultaneous ingestion from multiple sources with source tagging for traceability.

**Stage 2 - Data Profiling & Quality Scoring:** Comprehensive data profiling using Great Expectations and Pandas Profiling. Detect 15+ data quality issues: missing values, duplicates, outliers, type mismatches, inconsistent formats, whitespace issues, special characters, invalid dates, unexpected negative values, distribution anomalies, sparse columns, high cardinality issues, cross-column inconsistencies, range violations, and encoding issues. Calculate quality score (0-100) for each dataset based on issue severity and prevalence.

**Stage 3 - SourceGuard Multi-Source Reconciliation:** For multiple data sources: calculate reliability score for each source. Detect conflicts when the same entity has different values across sources. Apply intelligent conflict resolution using confidence scoring and source reliability ranking.

**Stage 4 - CleanQuery Module with Transparent Change Tracking:** Rule-based pipeline with type inference, missing value handling, outlier detection, format normalization, duplicate removal, and schema validation. Generate detailed change log showing before/after values, number of cells modified, rows removed, and types of changes applied.

**Stage 5 - InsightFlow Module with Impact Awareness:** Schema injection → LLM generates SQL → sqlglot validates → keyword blacklist check → column verification → read-only execution. Enhanced with impact analysis to identify potential downstream effects.

**Stage 6 - Interactive Insights & Visualization:** Natural language prompt interface for insights. System interprets intent, executes query, generates statistics, auto-selects visualization type, and renders interactive visualizations with contextual insights.

### 2.1.3 New Capabilities: Data Quality & Impact Analysis

- DataQualityScorer calculates 0-100 quality scores based on:
  - Completeness (25%): % of non-null values
  - Consistency (20%): % of uniform data types and formats
  - Accuracy (20%): % of values within expected ranges
  - Uniqueness (15%): % of unique records vs duplicates
  - Timeliness (10%): freshness of data
  - Validity (10%): % of records passing business rules
- Root Cause Analysis & Issue Localization provides:
  - Issue type and severity (Critical/High/Medium/Low)
  - Root cause hypothesis with supporting evidence
  - Affected data volume (rows and cells)
  - Exact location: row numbers, column names, cell addresses
  - Source origin and example records
- Impact Analysis Framework quantifies how data quality issues affect analytical outcomes (e.g., 'Duplicate records could overstate revenue by ~8%').
- Conclusion: DataTalk with multi-source support and quality assessment is TECHNICALLY FEASIBLE. Enhanced tech stack is proven. Six-stage pipeline is well-defined. Core challenges have clear solutions.

## 2.2 Economic Feasibility

### 2.2.1 Updated Budget Analysis

Total project cost ranges from 0 PKR (fully open source) to 15,000 PKR (with OpenAI API and additional features).

**Key cost components:**

- **Development:** Python, FastAPI, VS Code, Git → All Free.
- **Backend:** PostgreSQL, SQLAlchemy, Pandas, Great Expectations, Pandas Profiling → All Free.
- **Frontend:** React, Plotly, Altair → All Free.
- **Data Processing:** openpyxl, pyarrow, chardet, FuzzyWuzzy → All Free.
- **NLP/LLM:** OpenAI GPT-4 API (Optional) → 0-5,000 PKR.
- **Hosting:** AWS Free Tier EC2, Vercel → All Free
- **Conclusion:** DataTalk is ECONOMICALLY HIGHLY FEASIBLE. True minimum cost is 0 PKR. Optional OpenAI API adds up to 5,000-15,000 PKR. No licensing costs.

## 2.3 Operational Feasibility

The developer has strong Python and database skills with adequate knowledge of web development, LLM integration, and DevOps. Extended workload (750 hours over 7 months, ~24 hours/week average) is realistic.

**Timeline breakdown:**

- **Month 1:** Setup, Architecture, API Boilerplate (18 hrs/wk).
- **Month 2:** CleanQuery Module (26 hrs/wk).
- **Month 3:** Quality Profiling & Scoring (26 hrs/wk).
- **Month 4:** SourceGuard Multi-Source Reconciliation (26 hrs/wk).
- **Month 5:** InsightFlow NLP-to-SQL (26 hrs/wk).
- **Month 6:** Frontend & Insights Dashboard (26 hrs/wk).
- **Month 7:** Testing, Documentation, Deployment (22 hrs/wk).
- **Total:** ~750 hours over 7 months
- **Conclusion:** DataTalk is OPERATIONALLY FEASIBLE. Skills align. Workload is realistic. Risks are identifiable and mitigatable.

## 2.4 Legal and Regulatory Feasibility

**Key legal considerations:**

- **User Data Ownership:** All uploaded data remains user property.
- **Storage:** Auto-delete after 30 days.
- **Access Control:** Read-only database role.
- **API Privacy:** Only schema metadata sent to OpenAI, never actual data rows.
- **Pakistan Compliance:** No PDPA equivalent in force; academic exemptions apply
- **Conclusion:** DataTalk is LEGALLY FEASIBLE. No regulatory barriers for academic FYP. Standard data protection is adequate.

## 2.5 Schedule Feasibility

**Updated project phases:**

- **Phase 0:** Setup (1 week).
- **Phase 1:** Multi-Source Ingestion (2 weeks).
- **Phase 2:** CleanQuery Module (3 weeks).
- **Phase 3:** Quality Scoring & Profiling (3 weeks).
- **Phase 4:** SourceGuard Reconciliation (2 weeks).
- **Phase 5:** InsightFlow NLP-to-SQL (3 weeks).
- **Phase 6:** Frontend & Dashboard (3 weeks).
- **Phase 7:** Testing & Documentation (2 weeks).
- **Buffer: 2** weeks for adjustments
- **Total:** 21 weeks (fits 24-week FYP window)
- **Conclusion:** DataTalk is SCHEDULE FEASIBLE. 19-week critical path + 2-week buffer = 21 weeks, fitting the 24-week FYP window.

# 3. RESULTS AND DISCUSSION

## 3.1 Overall Feasibility Summary

**Technical:** HIGHLY FEASIBLE - Proven stack with Great Expectations, six-stage pipeline well-defined, all core challenges have clear solutions.
**Economic:** HIGHLY FEASIBLE - 0 PKR minimum, AWS Free Tier, no licensing costs, optional OpenAI adds 5K-15K PKR.
**Operational:** HIGHLY FEASIBLE - Skills strongly match, 750-hour workload over 7 months realistic, learning curve minimal.
**Legal:** FEASIBLE - No regulatory barriers for academic FYP, standard data protection adequate.
**Schedule:** HIGHLY FEASIBLE - 19-week critical path plus 2-week buffer fits 24-week window.

## 3.2 Key Success Factors

1. Strong data cleaning foundation (garbage in = garbage out)
2. Accurate quality scoring with domain expert calibration
3. Precise root cause analysis for data issues
4. Robust entity matching multi-source integration
5. Transparent change tracking and user trust
6. Robust schema injection into LLM
7. Comprehensive error messages and impact warnings
8. Regular testing with real, messy datasets
9. MVP scope discipline (defer complex features)
10. Early supervisor feedback on design decisions
11. Incremental deployment with user feedback

## 3.3 Major Enhancements Over Original Design

**1. Multi-Source Data Integration:** Support for simultaneous ingestion from multiple sources with source tagging and conflict detection.
**2. Comprehensive Quality Assessment:** Quality scoring (0-100) and detection of 15+ data quality issues with precise location identification.
**3. Root Cause Analysis:** System explains root causes, not just identifies issues (e.g., 'Missing values from legacy DB export')
**4. Transparent Change Tracking:** Detailed reports showing exactly what changed: rows removed, cells fixed, values imputed
**5. Impact Analysis:** Quantifies how data quality issues affect analytical outcomes (e.g., 'Duplicates overstate revenue by 8%').

# 4. CONCLUSION AND RECOMMENDATIONS

**PROCEED WITH DEVELOPMENT OF DATATALK**

DataTalk, enhanced with multi-source integration, comprehensive data quality assessment, root cause analysis, transparent change tracking, and impact analysis, is a highly feasible, high-impact Final Year Project addressing critical gaps in data accessibility and governance.
The project is technically sound, economically viable, operationally achievable, legally compliant, and schedulable within the FYP timeline. It demonstrates mastery of full-stack development, data engineering, ML integration, and user-centered design. The work has strong potential for job applications, graduate school consideration, and academic publication in data science venues.

**Recommended immediate next steps:**

1. Obtain supervisor approval on feasibility findings
2. Review data quality standards literature (ISO, DAMA, NIST)
3. Design detailed database schema for metadata
4. Create frontend wireframes
5. Set up development environment immediately
6. Begin Phase 0-1 implementation
7. Plan user testing strategy with real datasets8. Document progress weekly and solicit feedback